

ME46707 Computational Fluid Dynamics

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1 Unit-1: Introduction

- L-1: Basics of Computational Fluid Dynamics
- L-2: Governing Differential Equations and Boundary Conditions
- L-3: Mathematical Behaviour of Partial Differential Equations

After completion of this course, students will be able to

- 1 Define the governing differential equations and boundary conditions.
- 2 Determine the mathematical behaviour of partial differential equations.

Learning Objectives

After completion of this unit, students will be able to

- 1 Understand the basics of computational fluid dynamics.
- 2 Define the governing differential equations and boundary conditions.
- 3 Determine the mathematical behaviour of partial differential equations.

What is Computational Fluid Dynamics (CFD)?

- Computational Fluid Dynamics (CFD) is a branch of continuum mechanics that deals with the numerical simulation of fluid flow and heat transfer problems. OR
- Computational Fluid Dynamics (CFD) is the science of predicting fluid flow and heat transfer by numerically solving the governing equations of fluid mechanics using computers.
- CFD combines:
 - Fluid Mechanics
 - Heat Transfer
 - Numerical Methods
 - Computer Programming
 - High Performance Computing

Why Study CFD?

Experimental Analysis

- Expensive and often impossible
- Time consuming
- Prototype required
- Measurement errors
- Difficult visualization

CFD Analysis

- Cost effective
- Faster
- Virtual testing
- More complete information
- Easy optimization

Hence, CFD reduces design time and development cost.

Where is CFD?

- **Aerospace and Defence:** Air/Spacecraft, Avionics cooling
- **Automotive/mobile:** Engine Design, Battery cooling
- **Bio-medical:** Flow through artery and veins
- **Chemical Industry:** Mixing of chemical streams
- **Electronics:** Cooling of micro-circuits, Chip cooling
- **Energy:** Wing turbine
- **Fluid Structure Interaction:** Flexible membrane
- **I. C. Engines:** Combustion modelling in I.C. engines
- **Marine:** Flow of boats/ships
- **Material processing:** Grain growth, Butt welded joint
- **Micro-fluidics:** Droplet dynamics, Micro-reactor
- **Sports:** Driving a racing car
- **Turbo-machinery:** Design of blades of impulse turbine

How does a CFD Code Work?

A typical CFD code consists of three main components:

① **Pre-processor:**

- Define the geometry of the region of interest.
- Generate the computational mesh/grid.
- Select physical phenomena that need to be modelled.
- Define the properties of fluid and solid.
- Specify the initial and boundary conditions.

② **Solver:**

- Select the solution methods for pressure-velocity coupling.
- Specify the discretization techniques.
- Set the convergence criterion.
- Specify the initial value of variables.

③ **Post-processor:**

- Generate the contour and vector plots.
- Plot the line and 2D and 3D surface plots.
- Extract the post-script data file.

General Transport Equation

- The general transport equation for any transport property ϕ is:

$$\underbrace{\frac{\partial(\rho\phi)}{\partial t}}_{\text{Transient term}} + \underbrace{\nabla \cdot (\rho\phi \vec{V})}_{\text{Advection term}} = \underbrace{S_\phi}_{\text{Source term}} + \underbrace{\nabla \cdot (\Gamma \nabla \phi)}_{\text{Diffusion term}} \quad (1)$$

where ρ is density, $\vec{V}$ is velocity vector and Γ is diffusion coefficient.

- In Cartesian coordinates system:

$$\frac{\partial(\rho\phi)}{\partial t} + \frac{\partial(\rho u\phi)}{\partial x} + \frac{\partial(\rho v\phi)}{\partial y} + \frac{\partial(\rho w\phi)}{\partial z} = S_\phi + \frac{\partial^2(\Gamma\phi)}{\partial x^2} + \frac{\partial^2(\Gamma\phi)}{\partial y^2} + \frac{\partial^2(\Gamma\phi)}{\partial z^2} \quad (2)$$

- In Cylindrical polar coordinates system:

$$\begin{aligned} \frac{\partial(\rho\phi)}{\partial t} + \frac{1}{r} \frac{\partial(r\rho v_r\phi)}{\partial r} + \frac{1}{r} \frac{\partial(\rho v_\theta\phi)}{\partial \theta} + \frac{\partial(\rho v_z\phi)}{\partial z} \\ = S_\phi + \frac{1}{r} \frac{\partial}{\partial r} (r\Gamma \frac{\partial\phi}{\partial r}) + \\ \frac{1}{r^2} \frac{\partial}{\partial \theta} (\Gamma \frac{\partial\phi}{\partial \theta}) + \frac{\partial}{\partial z} (\Gamma \frac{\partial\phi}{\partial z}) \end{aligned}$$

(3)

Mass Transport Equation

- For Mass Transport Equation, $[\phi = 1, S_\phi = 0, \Gamma = 0]$
- Substituting into the general transport equation,

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \vec{V}) = 0 \quad (4)$$

- In Cartesian coordinates system:

$$\frac{\partial \rho}{\partial t} + \frac{\partial(\rho u)}{\partial x} + \frac{\partial(\rho v)}{\partial y} + \frac{\partial(\rho w)}{\partial z} = 0 \quad (5)$$

- In Cylindrical polar coordinates system:.,

$$\frac{\partial \rho}{\partial t} + \frac{1}{r} \frac{\partial(r \rho v_r)}{\partial r} + \frac{1}{r} \frac{\partial(\rho v_\theta)}{\partial \theta} + \frac{\partial(\rho v_z)}{\partial z} = 0 \quad (6)$$

Momentum Transport Equation

- For Momentum Transport Equation, $[\phi = \vec{V}, S_\phi = \rho \vec{g} - \nabla p, \Gamma = \mu]$
- Substituting into the general transport equation

$$\frac{\partial(\rho \vec{V})}{\partial t} + \nabla \cdot (\rho \vec{V} \vec{V}) = \rho \vec{g} - \nabla p + \nabla \cdot (\mu \nabla \vec{V})$$

$$\frac{\partial(\rho \vec{V})}{\partial t} + \nabla \cdot (\rho \vec{V} \vec{V}) = \rho \vec{g} - \nabla p + \mu \left[\nabla^2 \vec{V} + \frac{1}{3} \nabla(\nabla \cdot \vec{V}) \right] \quad (7)$$

- For incompressible flow,

$$\frac{\partial \vec{V}}{\partial t} + \nabla \cdot (\vec{V} \vec{V}) = \vec{g} - \frac{1}{\rho} \nabla p + \frac{\mu}{\rho} \left[\nabla^2 \vec{V} + \frac{1}{3} \nabla(\nabla \cdot \vec{V}) \right]$$

X-Momentum Transport Equation

$$\begin{aligned}
 & \frac{\partial(\rho u)}{\partial t} + \frac{\partial(\rho u u)}{\partial x} + \frac{\partial(\rho u v)}{\partial y} + \frac{\partial(\rho u w)}{\partial z} \\
 & = \rho g_x - \frac{\partial p}{\partial x} + \mu \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} \right) \\
 & + \mu \times \frac{1}{3} \frac{\partial}{\partial x} \left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} \right) \\
 & \frac{\partial(\rho u)}{\partial t} + u \frac{\partial(\rho u)}{\partial x} + v \frac{\partial(\rho u)}{\partial y} + w \frac{\partial(\rho u)}{\partial z} \\
 & = \rho g_x - \frac{\partial p}{\partial x} + \mu \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} \right) \\
 & + \mu \times \frac{1}{3} \frac{\partial}{\partial x} \left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} \right) \\
 & \frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} + w \frac{\partial u}{\partial z} \\
 & = g_x - \frac{1}{\rho} \frac{\partial p}{\partial x} + \frac{\mu}{\rho} \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} \right) \\
 & + \frac{\mu}{\rho} \times \frac{1}{3} \frac{\partial}{\partial x} \left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} \right)
 \end{aligned} \tag{8}$$

Y-Momentum Transport Equation

$$\begin{aligned}
 & \frac{\partial(\rho v)}{\partial t} + \frac{\partial(\rho uv)}{\partial x} + \frac{\partial(\rho vv)}{\partial y} + \frac{\partial(\rho vw)}{\partial z} \\
 & = \rho g_y - \frac{\partial p}{\partial y} + \mu \left(\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} + \frac{\partial^2 v}{\partial z^2} \right) \\
 & + \mu \times \frac{1}{3} \frac{\partial}{\partial y} \left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} \right) \\
 & \frac{\partial(\rho v)}{\partial t} + u \frac{\partial(\rho v)}{\partial x} + v \frac{\partial(\rho v)}{\partial y} + w \frac{\partial(\rho v)}{\partial z} \\
 & = \rho g_y - \frac{\partial p}{\partial y} + \mu \left(\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} + \frac{\partial^2 v}{\partial z^2} \right) \\
 & + \mu \times \frac{1}{3} \frac{\partial}{\partial y} \left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} \right) \\
 & \frac{\partial v}{\partial t} + u \frac{\partial v}{\partial x} + v \frac{\partial v}{\partial y} + w \frac{\partial v}{\partial z} \\
 & = g_y - \frac{1}{\rho} \frac{\partial p}{\partial y} + \frac{\mu}{\rho} \left(\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} + \frac{\partial^2 v}{\partial z^2} \right) \\
 & + \frac{\mu}{\rho} \times \frac{1}{3} \frac{\partial}{\partial y} \left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} \right)
 \end{aligned} \tag{9}$$

Z-Momentum Transport Equation

$$\begin{aligned}
 & \frac{\partial(\rho w)}{\partial t} + \frac{\partial(\rho u w)}{\partial x} + \frac{\partial(\rho v w)}{\partial y} + \frac{\partial(\rho w w)}{\partial z} \\
 &= \rho g_z - \frac{\partial p}{\partial z} + \mu \left(\frac{\partial^2 w}{\partial x^2} + \frac{\partial^2 w}{\partial y^2} + \frac{\partial^2 w}{\partial z^2} \right) \\
 &+ \mu \times \frac{1}{3} \frac{\partial}{\partial z} \left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} \right) \\
 & \frac{\partial(\rho w)}{\partial t} + u \frac{\partial(\rho w)}{\partial x} + v \frac{\partial(\rho w)}{\partial y} + w \frac{\partial(\rho w)}{\partial z} \\
 &= \rho g_z - \frac{\partial p}{\partial z} + \mu \left(\frac{\partial^2 w}{\partial x^2} + \frac{\partial^2 w}{\partial y^2} + \frac{\partial^2 w}{\partial z^2} \right) \\
 &+ \mu \times \frac{1}{3} \frac{\partial}{\partial z} \left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} \right) \\
 & \frac{\partial w}{\partial t} + u \frac{\partial w}{\partial x} + v \frac{\partial w}{\partial y} + w \frac{\partial w}{\partial z}
 \end{aligned} \tag{10}$$

$$\begin{aligned}
 &= g_z - \frac{1}{\rho} \frac{\partial p}{\partial z} + \frac{\mu}{\rho} \left(\frac{\partial^2 w}{\partial x^2} + \frac{\partial^2 w}{\partial y^2} + \frac{\partial^2 w}{\partial z^2} \right) \\
 &+ \frac{\mu}{\rho} \times \frac{1}{3} \frac{\partial}{\partial z} \left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} \right)
 \end{aligned}$$

Radial (r)-Momentum Transport Equation

$$\begin{aligned}
 & \frac{\partial(\rho v_r)}{\partial t} + \frac{v_r}{r} \frac{\partial(r \rho v_r)}{\partial r} + \frac{v_\theta}{r} \frac{\partial(\rho v_r)}{\partial \theta} \\
 & - \frac{\rho v_\theta^2}{r} + v_z \frac{\partial(\rho v_r)}{\partial z} = \rho g_r - \frac{\partial p}{\partial r} \\
 & + \mu \left[\frac{\partial}{\partial r} \left(\frac{1}{r} \frac{\partial(r v_r)}{\partial r} \right) + \frac{1}{r^2} \frac{\partial^2 v_r}{\partial \theta^2} + \frac{\partial^2 v_r}{\partial z^2} - \frac{2}{r^2} \frac{\partial v_\theta}{\partial \theta} \right] \\
 & + \mu \times \frac{1}{3} \frac{\partial}{\partial r} \left[\frac{1}{r} \frac{\partial(r v_r)}{\partial r} + \frac{1}{r} \frac{\partial v_\theta}{\partial \theta} + \frac{\partial v_z}{\partial z} \right] \\
 \\
 & \frac{\partial(v_r)}{\partial t} + \frac{v_r}{r} \frac{\partial(r v_r)}{\partial r} + \frac{v_\theta}{r} \frac{\partial(v_r)}{\partial \theta} \\
 & - \frac{v_\theta^2}{r} + v_z \frac{\partial(v_r)}{\partial z} = g_r - \frac{1}{\rho} \frac{\partial p}{\partial r} \\
 & + \frac{\mu}{\rho} \left[\frac{\partial}{\partial r} \left(\frac{1}{r} \frac{\partial(r v_r)}{\partial r} \right) + \frac{1}{r^2} \frac{\partial^2 v_r}{\partial \theta^2} + \frac{\partial^2 v_r}{\partial z^2} - \frac{2}{r^2} \frac{\partial v_\theta}{\partial \theta} \right] \\
 & + \frac{1}{3} \times \frac{\mu}{\rho} \frac{\partial}{\partial r} \left[\frac{1}{r} \frac{\partial(r v_r)}{\partial r} + \frac{1}{r} \frac{\partial v_\theta}{\partial \theta} + \frac{\partial v_z}{\partial z} \right]
 \end{aligned}$$

(11)

Tangential (θ)-Momentum Transport Equation

$$\begin{aligned}
 \frac{\partial(\rho v_\theta)}{\partial t} + \frac{v_r}{r} \frac{\partial(r \rho v_\theta)}{\partial r} + \frac{v_\theta}{r} \frac{\partial(\rho v_\theta)}{\partial \theta} \\
 + \frac{v_r v_\theta}{r} + v_z \frac{\partial(\rho v_\theta)}{\partial z} = \rho g_\theta - \frac{1}{r} \frac{\partial p}{\partial \theta} \\
 + \mu \left[\frac{\partial}{\partial r} \left(\frac{1}{r} \frac{\partial(r v_\theta)}{\partial r} \right) + \frac{1}{r^2} \frac{\partial^2 v_\theta}{\partial \theta^2} + \frac{\partial^2 v_\theta}{\partial z^2} + \frac{2}{r^2} \frac{\partial v_r}{\partial \theta} \right] \\
 + \mu \times \frac{1}{3} \times \frac{1}{r} \frac{\partial}{\partial \theta} \left[\frac{1}{r} \frac{\partial(r v_r)}{\partial r} + \frac{1}{r} \frac{\partial v_\theta}{\partial \theta} + \frac{\partial v_z}{\partial z} \right]
 \end{aligned}$$

$$\begin{aligned}
 \frac{\partial v_\theta}{\partial t} + \frac{v_r}{r} \frac{\partial(r v_\theta)}{\partial r} + \frac{v_\theta}{r} \frac{\partial v_\theta}{\partial \theta} \\
 + \frac{v_r v_\theta}{r} + v_z \frac{\partial v_\theta}{\partial z} = g_\theta - \frac{1}{\rho r} \frac{\partial p}{\partial \theta} \\
 + \frac{\mu}{\rho} \left[\frac{\partial}{\partial r} \left(\frac{1}{r} \frac{\partial(r v_\theta)}{\partial r} \right) + \frac{1}{r^2} \frac{\partial^2 v_\theta}{\partial \theta^2} + \frac{\partial^2 v_\theta}{\partial z^2} + \frac{2}{r^2} \frac{\partial v_r}{\partial \theta} \right] \\
 + \frac{1}{3} \times \frac{\mu}{\rho} \times \frac{1}{r} \frac{\partial}{\partial \theta} \left[\frac{1}{r} \frac{\partial(r v_r)}{\partial r} + \frac{1}{r} \frac{\partial v_\theta}{\partial \theta} + \frac{\partial v_z}{\partial z} \right]
 \end{aligned}$$

(12)

Axial (z)-Momentum Transport Equation

$$\begin{aligned}
 \frac{\partial(\rho v_z)}{\partial t} + \frac{v_r}{r} \frac{\partial(r \rho v_z)}{\partial r} + \frac{v_\theta}{r} \frac{\partial(\rho v_z)}{\partial \theta} + v_z \frac{\partial(\rho v_z)}{\partial z} \\
 = \rho g_z - \frac{\partial p}{\partial z} + \\
 \mu \left[\frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial v_z}{\partial r} \right) + \frac{1}{r^2} \frac{\partial^2 v_z}{\partial \theta^2} + \frac{\partial^2 v_z}{\partial z^2} \right] + \\
 \mu \times \frac{1}{3} \frac{\partial}{\partial z} \left[\frac{1}{r} \frac{\partial(r v_r)}{\partial r} + \frac{1}{r} \frac{\partial v_\theta}{\partial \theta} + \frac{\partial v_z}{\partial z} \right] \\
 \frac{\partial v_z}{\partial t} + \frac{v_r}{r} \frac{\partial(r v_z)}{\partial r} + \frac{v_\theta}{r} \frac{\partial v_z}{\partial \theta} + v_z \frac{\partial v_z}{\partial z} \\
 = g_z - \frac{1}{\rho} \frac{\partial p}{\partial z} \\
 + \frac{\mu}{\rho} \left[\frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial v_z}{\partial r} \right) + \frac{1}{r^2} \frac{\partial^2 v_z}{\partial \theta^2} + \frac{\partial^2 v_z}{\partial z^2} \right] \\
 + \frac{1}{3} \times \frac{\mu}{\rho} \times \frac{\partial}{\partial z} \left[\frac{1}{r} \frac{\partial(r v_r)}{\partial r} + \frac{1}{r} \frac{\partial v_\theta}{\partial \theta} + \frac{\partial v_z}{\partial z} \right]
 \end{aligned}
 \tag{13}$$

Energy Transport Equation

- For Energy Transport Equation, $[\phi = e, S_\phi = \dot{q}''', \Gamma = k/C_p]$
where $e = C_p T$

- Substituting into the general transport equation

$$\frac{\partial(\rho C_p T)}{\partial t} + \nabla \cdot (\rho C_p T \vec{V}) = \dot{q}''' + \nabla \cdot (k \nabla T) \quad (14)$$

- In Cartesian Coordinates system:

$$\begin{aligned} \frac{\partial(\rho C_p T)}{\partial t} + \frac{\partial(\rho C_p u T)}{\partial x} + \frac{\partial(\rho C_p v T)}{\partial y} + \frac{\partial(\rho C_p w T)}{\partial z} \\ = \dot{q}''' + \frac{\partial}{\partial x} \left(k \frac{\partial T}{\partial x} \right) \\ + \frac{\partial}{\partial y} \left(k \frac{\partial T}{\partial y} \right) + \frac{\partial}{\partial z} \left(k \frac{\partial T}{\partial z} \right) \\ \frac{\partial(\rho C_p T)}{\partial t} + u \frac{\partial(\rho C_p T)}{\partial x} + v \frac{\partial(\rho C_p T)}{\partial y} + w \frac{\partial(\rho C_p T)}{\partial z} \\ = \dot{q}''' + \frac{\partial}{\partial x} \left(k \frac{\partial T}{\partial x} \right) \\ + \frac{\partial}{\partial y} \left(k \frac{\partial T}{\partial y} \right) + \frac{\partial}{\partial z} \left(k \frac{\partial T}{\partial z} \right) \end{aligned}$$

Energy Transport Equation, Contd...

- In Cylindrical polar coordinates system:

$$\begin{aligned} \frac{\partial(\rho C_p T)}{\partial t} + \frac{\partial(\rho C_p v_r T)}{\partial r} + \frac{1}{r} \frac{\partial(\rho C_p v_\theta T)}{\partial \theta} \\ + \frac{\partial(\rho C_p v_z T)}{\partial z} = \dot{q}''' + \frac{1}{r} \frac{\partial}{\partial r} \left(r k \frac{\partial T}{\partial r} \right) \\ + \frac{1}{r^2} \frac{\partial}{\partial \theta} \left(k \frac{\partial T}{\partial \theta} \right) + \frac{\partial}{\partial z} \left(k \frac{\partial T}{\partial z} \right) \end{aligned}$$
$$\begin{aligned} \frac{\partial(\rho C_p T)}{\partial t} + v_r \frac{\partial(\rho C_p T)}{\partial r} + \frac{v_\theta}{r} \frac{\partial(\rho C_p T)}{\partial \theta} \\ + v_z \frac{\partial(\rho C_p T)}{\partial z} = \dot{q}''' + \frac{1}{r} \frac{\partial}{\partial r} \left(r k \frac{\partial T}{\partial r} \right) \\ + \frac{1}{r^2} \frac{\partial}{\partial \theta} \left(k \frac{\partial T}{\partial \theta} \right) + \frac{\partial}{\partial z} \left(k \frac{\partial T}{\partial z} \right) \end{aligned} \quad (16)$$

Chemical Species Transport Equation

- For Chemical Species Transport Equation,
[$\phi = Y_i, S_\phi = R_i + S_i, \Gamma = \rho D$]
- Substituting into the general transport equation

$$\frac{\partial(\rho Y_i)}{\partial t} + \nabla \cdot (\rho Y_i \vec{V}) = R_i + S_i + \nabla \cdot (\rho D \nabla Y_i) \quad (17)$$

- In Cartesian coordinates system:

$$\begin{aligned} \frac{\partial(\rho Y_i)}{\partial t} + \frac{\partial(\rho u Y_i)}{\partial x} + \frac{\partial(\rho v Y_i)}{\partial y} + \frac{\partial(\rho w Y_i)}{\partial z} \\ = R_i + S_i + \frac{\partial}{\partial x}(\rho D \frac{\partial Y_i}{\partial x}) \\ + \frac{\partial}{\partial y}(\rho D \frac{\partial Y_i}{\partial y}) + \frac{\partial}{\partial z}(\rho D \frac{\partial Y_i}{\partial z}) \\ \frac{\partial(\rho Y_i)}{\partial t} + u \frac{\partial(\rho Y_i)}{\partial x} + v \frac{\partial(\rho Y_i)}{\partial y} + w \frac{\partial(\rho Y_i)}{\partial z} \\ = R_i + S_i + \frac{\partial}{\partial x}(\rho D \frac{\partial Y_i}{\partial x}) \\ + \frac{\partial}{\partial y}(\rho D \frac{\partial Y_i}{\partial y}) + \frac{\partial}{\partial z}(\rho D \frac{\partial Y_i}{\partial z}) \end{aligned}$$

Chemical Species Transport Equation, Contd...

- In Cylindrical polar coordinates system:

$$\begin{aligned} \frac{\partial(\rho Y_i)}{\partial t} + \frac{1}{r} \frac{\partial(r \rho v_r Y_i)}{\partial r} + \frac{1}{r} \frac{\partial(\rho v_\theta Y_i)}{\partial \theta} \\ + \frac{\partial(\rho v_z Y_i)}{\partial z} = R_i + S_i + \frac{1}{r} \frac{\partial}{\partial r} (r \rho D \frac{\partial Y_i}{\partial r}) \\ + \frac{1}{r^2} \frac{\partial}{\partial \theta} (\rho D \frac{\partial Y_i}{\partial \theta}) + \frac{\partial}{\partial z} (\rho D \frac{\partial Y_i}{\partial z}) \end{aligned}$$
$$\begin{aligned} \frac{\partial(\rho Y_i)}{\partial t} + \frac{v_r}{r} \frac{\partial(r \rho Y_i)}{\partial r} + \frac{v_\theta}{r} \frac{\partial(\rho Y_i)}{\partial \theta} \\ + v_z \frac{\partial(\rho Y_i)}{\partial z} = R_i + S_i + \frac{1}{r} \frac{\partial}{\partial r} (r \rho D \frac{\partial Y_i}{\partial r}) \\ + \frac{1}{r^2} \frac{\partial}{\partial \theta} (\rho D \frac{\partial Y_i}{\partial \theta}) + \frac{\partial}{\partial z} (\rho D \frac{\partial Y_i}{\partial z}) \end{aligned} \quad (19)$$

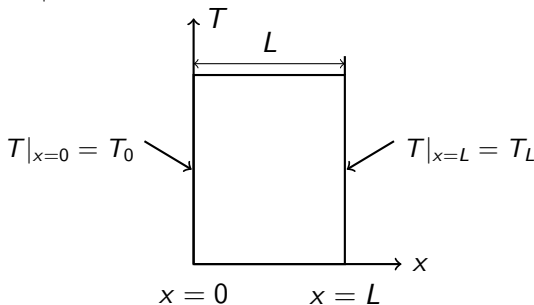
Representation of the General Transport Equation

Table : Representation of the General Transport Equation

Transport Equations	Transport Property (ϕ)	Source term (S_ϕ)	Diffusion Coefficient (Γ)
Mass	1	0	0
Momentum	$\vec{V}$	$\rho \vec{g} - \nabla p$	μ
Energy	e $e = C_p T$	$\dot{q}'''$	k/C_p
Chemical Species	Y_i	$R_i + S_i$	ρD

BCs-1: Dirichlet Boundary Condition

- **Dirichlet Boundary Condition:** Temperature specified boundary condition
- Consider 1-D conduction through a plane wall of thickness L :
 $0 \leq x \leq L$
- **Boundary conditions:**
 - (i) At $x = 0$, $T|_{x=0} = T_0$
 - (ii) At $x = L$, $T|_{x=L} = T_L$



BCs-2: Neumann Boundary Condition

- **Neumann Boundary Condition:** Heat flux specified BCs

- **Case-I:** Finite heat flux specified BCs:

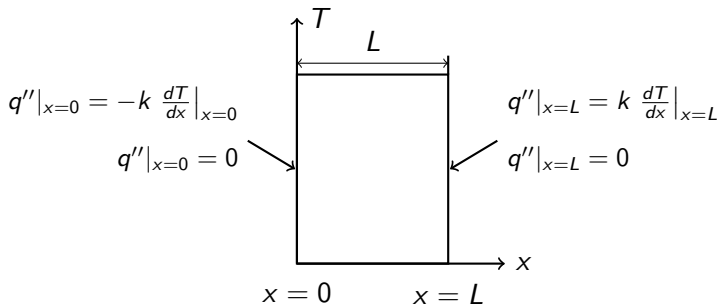
(i) At $x = 0$: $q''|_{x=0} = -k \frac{dT}{dx}|_{x=0}$

(ii) At $x = L$: $q''|_{x=L} = k \frac{dT}{dx}|_{x=L}$

- **Case-II:** Adiabatic/Insulated surface BCs:

(i) At $x = 0$: $-k \frac{dT}{dx}|_{x=0} = 0, \Rightarrow \frac{dT}{dx}|_{x=0} = 0$

(ii) At $x = L$: $k \frac{dT}{dx}|_{x=L} = 0, \Rightarrow \frac{dT}{dx}|_{x=L} = 0$

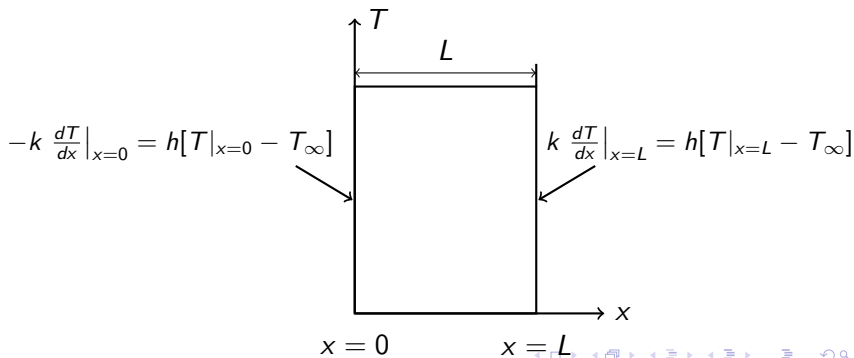


BCs-3: Robin Boundary Condition

- **Robin Boundary Condition:** Convection heat transfer specified Bcs
- Consider one-dimensional conduction through a plane wall exposed to fluids at both surfaces.

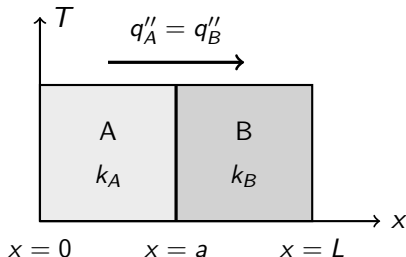
(i) At $x = 0$: $-k \frac{dT}{dx} \Big|_{x=0} = h[T|_{x=0} - T_{\infty}]$

(ii) At $x = L$: $k \frac{dT}{dx} \Big|_{x=L} = h[T|_{x=L} - T_{\infty}]$



BCs-4: Interface Boundary Condition

- Consider two materials joined at an interface:



- For two materials having equal thickness:

(i) **Continuity of Temperature:** $T_A = T_B$

$$\text{At interface } x = a \quad T|_{x=a^-} = T|_{x=a^+} \Rightarrow T_A|_{x=a} = T_B|_{x=a}$$

(ii) **Continuity of Heat Flux:** $q''_A = q''_B$

$$k_A \left. \frac{dT}{dx} \right|_{x=a^-} = k_B \left. \frac{dT}{dx} \right|_{x=a^+} \Rightarrow k_A \left. \frac{dT}{dx} \right|_{x=a} = k_B \left. \frac{dT}{dx} \right|_{x=a} = k_{\text{eff}} \left(\frac{dT}{dx} \right)$$

- The effective thermal conductivity is determined as:

$$\frac{1}{k_{\text{eff}}} = \frac{1}{2} \left[\frac{1}{k_A} + \frac{1}{k_B} \right] \Rightarrow k_{\text{eff}} = \frac{2k_A k_B}{k_A + k_B}$$

Mathematical Behaviour of PDEs

- Consider general second-order partial differential equation (PDE) in 2-D domain is written as:

$$A \frac{\partial^2 \phi}{\partial x^2} + B \frac{\partial^2 \phi}{\partial x \partial y} + C \frac{\partial^2 \phi}{\partial y^2} + D \frac{\partial \phi}{\partial x} + E \frac{\partial \phi}{\partial y} + F \phi + G = 0 \quad (20)$$

where the coefficients A, B, C, D, E, F and G are constants or may be functions of both the independent and/or dependent variables x, y .

- For the continuity of the first order derivatives of ϕ ,

$$d\phi_x = \frac{\partial \phi_x}{\partial x} dx + \frac{\partial \phi_x}{\partial y} dy = \frac{\partial^2 \phi}{\partial x^2} dx + \frac{\partial^2 \phi}{\partial x \partial y} dy \quad (21)$$

$$d\phi_y = \frac{\partial \phi_y}{\partial x} dx + \frac{\partial \phi_y}{\partial y} dy = \frac{\partial^2 \phi}{\partial x \partial y} dx + \frac{\partial^2 \phi}{\partial y^2} dy \quad (22)$$

- The equation (20), (21) and (22) can be written in matrix form as:

$$\begin{bmatrix} A & B & C \\ dx & dy & 0 \\ 0 & dx & dy \end{bmatrix} \begin{bmatrix} \phi_{xx} \\ \phi_{xy} \\ \phi_{yy} \end{bmatrix} = \begin{bmatrix} -H \\ d\phi_x \\ d\phi_y \end{bmatrix}$$

Derivation of Discriminant, $B^2 - 4AC$

- For discontinuities in the second-order derivatives of the dependent variables ($\phi_{xx}, \phi_{xy}, \phi_{yy}$) determinant of the coefficient matrix is equal to zero.

$$\begin{vmatrix} A & B & C \\ dx & dy & 0 \\ 0 & dx & dy \end{vmatrix} = 0$$

- The characteristic equation is written as:

$$A \left(\frac{dy}{dx} \right)^2 - B \left(\frac{dy}{dx} \right) + C = 0 \quad (23)$$

- For solving this quadratic equation yields the equation of the characteristics in physical space.

$$\frac{dy}{dx} = \frac{B \pm \sqrt{B^2 - 4AC}}{2A} \quad (24)$$

- Depending on the value of discriminant ($D = B^2 - 4AC$) characteristic curves can be real or imaginary.

Classification of PDEs

- The second-order PDE is classified according to the value of the discriminant ($D = B^2 - 4AC$) as given in table-1.

Table : Classification of second-order PDEs

Cases	Discriminant (D)	Equation Type	Characteristics
I	$B^2 - 4AC < 0$	Elliptic PDE Laplace equation $\frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} = 0$	0
II	$B^2 - 4AC = 0$	Parabolic PDE Heat diffusion equation $\frac{\partial \phi}{\partial t} = \alpha \frac{\partial^2 \phi}{\partial x^2}$	1
III	$B^2 - 4AC > 0$	Hyperbolic PDE Wave equation $\frac{\partial^2 \phi}{\partial t^2} = c^2 \frac{\partial^2 \phi}{\partial x^2}$	2

Nature of the Characteristics of PDEs

- Consider general second-order partial differential equation in 2-D domain is written as:

$$A \frac{\partial^2 \phi}{\partial x_1^2} + B \frac{\partial^2 \phi}{\partial x_1 \partial x_2} + C \frac{\partial^2 \phi}{\partial x_2^2} + H = 0 \quad (25)$$

- In general form for N independent variables $(x_1, x_2, \dots, x_N)$ is written as:

$$\sum_{i=1}^N \sum_{j=1}^N A_{ij} \frac{\partial^2 \phi}{\partial x_i \partial x_j} + H = 0 \quad (26)$$

where the coefficient matrix is symmetric, $A_{ij} = A_{ji}$ and H contains first-order derivative terms, and/or source terms.

- The coefficient matrix (A_{ij}) is written as:

$$\mathbf{A}_{ij} = \begin{bmatrix} A_{11} & A_{12} & A_{13} & \cdots & A_{1N} \\ A_{21} & A_{22} & A_{23} & \cdots & A_{2N} \\ A_{31} & A_{32} & A_{33} & \cdots & A_{3N} \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ A_{N1} & A_{N2} & A_{N3} & \cdots & A_{NN} \end{bmatrix}$$

Eigenvalue Criterion

- **Fletcher (1991)** explains that the nature of the characteristics of PDE can be determined on the basis of the eigenvalues of the coefficient matrix (A_{ij}).
- Hence, the eigenvalues are obtained from

$$\det(A_{ij} - \lambda I) = 0$$

$$\begin{vmatrix} A_{11} - \lambda & A_{12} & A_{13} & \cdots & A_{1N} \\ A_{21} & A_{22} - \lambda & A_{23} & \cdots & A_{2N} \\ A_{31} & A_{32} & A_{33} - \lambda & \cdots & A_{3N} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ A_{N1} & A_{N2} & A_{N3} & \cdots & A_{NN} - \lambda \end{vmatrix} = 0$$

- Hence, the sign of the eigenvalues of coefficient matrix determine the nature of the PDE.

Classification of PDEs

- The second-order PDE can be classified on the basis of the eigenvalue of the coefficient matrix as given in table-2.

Table : Classification of second-order PDEs

Cases	Eigenvalue (λ)	Equation Type	Characteristics
I	All eigenvalues $\lambda \neq 0$ and have the same sign	Elliptic PDE Laplace equation $\frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} = 0$	0
II	At least one eigenvalue $\lambda = 0$	Parabolic PDE Heat diffusion equation $\frac{\partial \phi}{\partial t} = \alpha \frac{\partial^2 \phi}{\partial x^2}$	1
III	All eigenvalues $\lambda \neq 0$ and one λ has opposite sign	Hyperbolic PDE Wave equation $\frac{\partial^2 \phi}{\partial t^2} = c^2 \frac{\partial^2 \phi}{\partial x^2}$	2